

Sunidhi Prakash

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EDUCATION

B.Tech, Indian Institute of Technology Guwahati, India <i>Bachelors in Engineering Physics</i> Course Work: Probability & Statistics, Computational Physics, Simulation Techniques, Machine Learning, Signals & Systems	<i>Jul 2021 – Jun 2025</i>
M.Sc., World Quant University <i>Masters in Financial Engineering</i> Course Work: Financial Markets, Financial Data, Financial Econometrics, Derivative Pricing, Stochastic Modelling	<i>July 2025 – Present</i>

EXPERIENCE

Axis Bank <i>Business Analyst - Retail & Deposit</i> - Built and deployed customer segmentation models using K-Means clustering & classification to optimize customer call scheduling, boosting cross-sell performance by 2%. - Built and shipped PySpark-based ML pipelines powering personalized outbound campaigns, with real-time eligibility scoring automated via Jenkins & Bitbucket CI/CD. - Led full model lifecycle - monitoring, backtesting, iterating performance with SQL & Python across 7+ variants in production.	June 2025 - Present Mumbai, India
Indian Institute of Technology, Guwahati <i>Research Intern under Dr. Sanasam Ranbir Singh in OSINT Lab</i> - Researched NLP & multimodal LLMs (LLaVA, MiniGPT-4, LLaMA-Adapter, InstructBLIP) for sarcasm detection across a labeled dataset of 12,000 social media posts combining text & images. Fine-tuned 4+ models on real-world data, designing evaluation splits & training workflows to preserve top-performing weights. - Benchmarked pretrained & fine-tuned models with precision/recall metrics to find performance gaps across data categories.	May 2024 - Sep 2024 Guwahati, India

PROJECTS

News-Driven Market Risk Indicator Using NLP <i>Personal Project</i> - Built an end-to-end forecasting pipeline integrating 5.6 years of S&P 500 price data with 879K+ GDELT news articles into ARIMA/ARIMAX (return) and GARCH/GARCH-X (volatility) models. - Validated performance via walk-forward backtesting with Diebold-Mariano significance testing across 639 out-of-sample days, and automated the pipeline to self update daily, re-fitting weekly on a rolling 3-year window.	Jan. 2026 - Feb. 2026 Github
Large-Scale Optimization via Sifting <i>IISC Bangalore, CiSTUP</i> - Engineered a column-generation (sifting) algorithm in Python/CPLEX to solve a 5,000-variable, 500-constraint LP, overcoming CPLEX Community Edition's 1,000-variable cap by iteratively pricing and solving reduced subproblems. - Converged the working variable set to 896 columns (82% reduction) in 8 iterations, achieving the global optimum with 0 constraint violations at 1e-9 tolerance.	Feb. 2024 - March. 2024 Github
Prompt Engineering & Evaluation Pipeline for LLMs <i>Guide: Prof. GS Setlur, Department of Physics, IIT Guwahati</i> - Built an automated evaluation pipeline across ChatGPT, Gemini, and LLaMA APIs using the LangChain framework (prompts, memory, chains, agents) to solve JEE Advanced-level quantitative problems. - Evaluated LLMs & designed prompts to analyze model accuracy across question types and identify systematic reasoning gaps.	Jan. 2024 - April 2024 Github
Multi-Resolution STFT Analysis <i>IIT Guwahati: Course Project</i> - Developed a multi-window STFT approach in Python to capture both transient and sustained audio features, overcoming time-frequency trade-offs - Tested on tonal and percussive sounds, showing improved temporal and spectral representation over Constant-Q Transform.	April. 2025 - May. 2025 Github

TECHNICAL SKILLS

- Statistics & Modelling:** Time Series Forecasting, Regression, Hypothesis Testing, Stochastic Processes
- Mathematical Methods:** Linear Algebra, Optimization, Numerical Techniques, Monte Carlo Simulation
- Programming:** C/C++, Python, R
- ML Frameworks:** Transformer, Hugging Face, Tensorflow
- Data & Tools:** Pandas, NumPy, SQL (PostgreSQL, MySQL)

ACHIEVEMENTS & CERTIFICATIONS

- Codeforces:** Rating: **1787** | **Expert** | **._flash_.** 🏆 | Round 988: Secured Global Rank 68 among 30k participants.
- Inter IIT:** Won **silver** in "Automated Research Paper Categorization" using **NLP** techniques and **Transformer** models.
- Stanford, Coursera:** Algorithms, PostgreSQL, Bayesian Statistics & Time Series Analysis, Machine Learning & Deep Learning